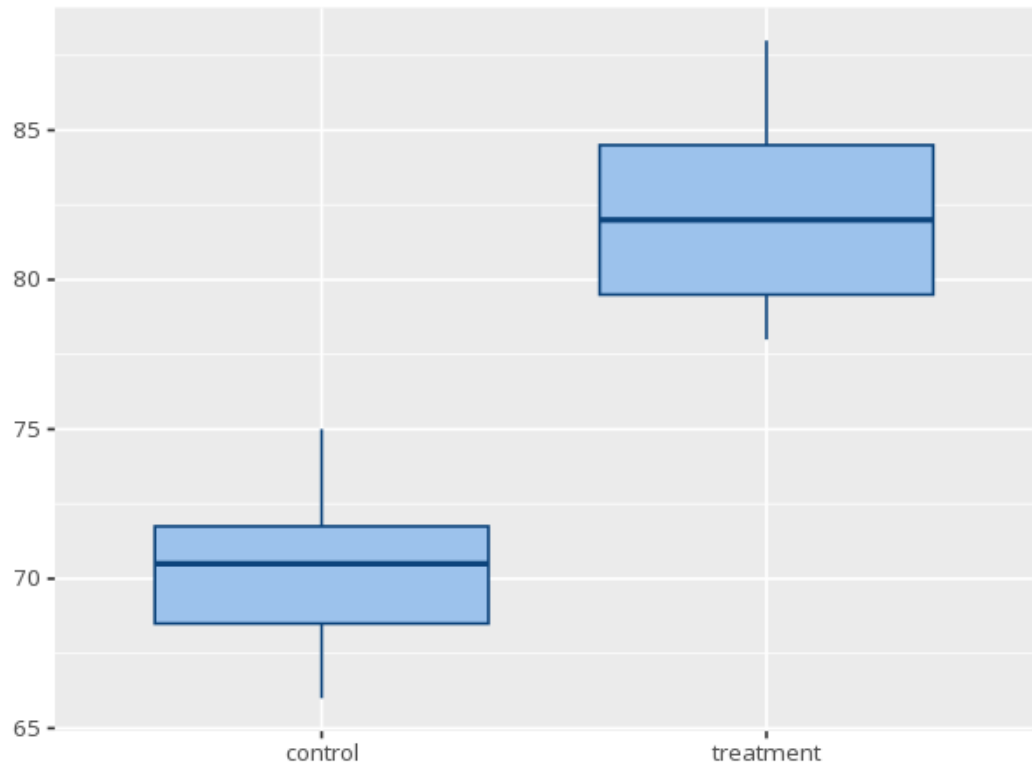


1 Independent t-test

Group	N	Mean	Std. Dev.	SE
control	6	70.33	3.14	1.28
treatment	6	82.33	3.78	1.54

Contrast	t	df	p	Mdiff	95% CI	Cohen's d [95% CI]
control – treatment	–5.98	9.68	<.001	–12.00	[–16.49, –7.51]	–3.45 [–5.32, –1.53]



Asks whether two groups' averages differ by more than chance. Look at p: below alpha (e.g. 0.05) means significant. The mean difference and 95% CI show the size and precision of the gap; Cohen's d gives effect size (~0.2 small, 0.5 medium, 0.8 large). A large p means no significant difference was detected — not that the group means are equal.

An independent-samples t-test compared control (M=70.3, SD=3.1) and treatment (M=82.3, SD=3.8), $t(9.68)=-5.98$, $p < .001$, $d=-3.45$ [–5.32, –1.53].

Statistical basis: Independent-samples t-test (pooled variance) (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Welch's correction (unequal variances, the app's default) (Welch, B. L. (1947). "The generalization of Student's problem when several different population variances are involved." *Biometrika*, 34(1-2), 28-35.); Cohen's d effect-size benchmarks (.2/.5/.8) (Co-

hen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." *Journal of the American Statistical Association*, 69(346), 364-367.).